Complex System Approaches: Could They Enhance the Relevance of Clinical Research?

To the Editor:

In his invited commentary on the article by Miller et al titled "A Quantitative Analysis of Research Publications in Physical Therapy Journals" in the February 2003 issue of *Physical Therapy*, Jette posed a simple question: "What can be done to stimulate more research in physical therapy that has direct clinical relevance?"[1(p132)] Jette proposed that, to answer this question, clinicians and scientists in physical therapy must come together. In fact, he proposed that the divergence between clinicians and scientists has "hindered the conduct of clinically relevant research that will meet the contemporary demands of society for clinical interventions rooted in sound theory and scientific evidence, versus tradition and anecdote."[1(p132)]

One etiology of the hindered conduct of clinically relevant research might be what we would call a "confounding factor" in the divide between clinicians and scientists. A confounding factor is a third factor, independently related to cause and effect. Here, *cause* is the divergence between clinicians and scientists; *effect* is hindrance of the conduct of clinically relevant research; and, I contend, a confounding factor is the complex system inherent in physical therapist practice. The presence of this complex system can independently lead to a divergence and to a hindrance. If left unstated and unmanaged, this *confounding factor* could lead to perpetual difficulties in remedying the problem concerning clinically relevant research.

The study of complex systems is now recognized as a scientific discipline.[2] The field is not foreign to physical therapy, having several ties to the disciplines of neurophysiology with various dynamical systems theories. What would be new, however, is the application of these principles to clinical research.

A complex system consists of interconnected parts. The simple study of the parts in isolation—reductionism—is the modus operandi of the scientific method, attempting to isolate sources of variation. Clinicians, however, are faced with *all* sources of variation at the same time and must deal constantly with the full burden of the complex system. They and their patient management become immersed, interconnected, and part of the complex system. This is a potential source of divergence between clinicians and scientists. Clinicians are fully aware that the patient, the components of patient management, the interaction between therapist and patient, and other factors form the entire system that ultimately determines outcome. This is, perhaps, a greater reality in physical therapy than in medicine, based on the level of inquiry that is influenced by the larger number of variables as we move from the cellular level to the level of human being—environmental and social interaction.

Scientists are aware of the dynamic interconnected relationships, but they respond by reducing complexity with study criteria and methods (reductionism) that contribute to the divergence. When I give a presentation on the impact of physical therapy—more specifically, exercise—on congestive heart failure, I provide the exclusion criteria for studies on effectiveness, and many physical therapists (from acute rehabilitation settings, skilled nursing facilities, and home care settings) look bewildered and tell me that I have just excluded all of their patients!

The presence of complex systems is inherent in physical therapist practice, and the inability of scientists to deal with this issue using standard scientific methods might be a source of *divergence* in clinician-to-scientist communication. Complex systems also can lead to a *hindrance* in conduct of clinically relevant research. Understanding component parts by studying them in isolation will never lead to a complete understanding of the complex system: the whole is more than the sum of its parts.

The very nature of clinically relevant research is to provide an understanding of how physical therapy—the entire activity of physical therapy—can lead to reduction of disability, improvement in function, reduction of impairment, prevention of

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pathology, and so on. By encapsulating the entire activity of physical therapy without an attempt to understand the nature of the complex system, we may end up using 2 equally problematic approaches. One approach is to reduce by eliminating factors (reductionism); the other approach is to reduce by aggregating multiple factors and assuming that the constitution of these aggregations is irrelevant to the research question. The use of complex systems in clinical research can take a variety of approaches and, in my opinion, will never replace commonplace reductionism, because there is still a role for understanding simple isolated relationships. My suggestion is to put these isolated relationships into a full context, appreciating that physical therapist practice is a complex system and that methods exist for development in this area.

I ask scientists, clinicians, and clinician scientists to consider a dialogue regarding complex systems as applied to clinical research in physical therapy. Inclusion of complex systems approaches to theory, study design, methods, conduct, and analysis may improve dialogue and enhance the clinical relevance of research in physical therapy, based on the premise that physical therapist clinical practice is a complex system.

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1 Jette AM. Invited commentary on "A quantitative analysis of research publications in physical therapy journals." *Phys Ther.* 2003;83:131-132.

2 Bar-Yam Y. *Dynamics of Complex Systems.* Reading, Mass: Perseus Books; 1997.

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